EP8199 - GLOBAL OPTIMIZATION OF MIXED-INTEGER AND NON-CONVEX PROGRAMS

COLLOQUIUM 4

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Outline

- RECAP:
- (NON-CONVEX) GENERALIZED BENDERS DECOMPOSITION

Procedure for Global Optimization I

The general procedure is to determine a sequence of **Lower Bounds LBDs** and **Upper Bounds UBDs** that bracket the optimal value. Converge the best known UBD and LBD such that the gap **UBD - LBD** is within a tolerance. Solvers usually report this gap.

Upper Bounding Techniques:

1. Determine any **feasible** point. Value of objective provides an UBD.
2. **Restriction**: Shrink the feasible set of an optimization problem by constraining it further. The optimal value or value or objective function value at any feasible point yield an UBD for the original optimization problem.

Lower Bounding Techniques:

1. **Relaxation** of constraints
2. **Convexification**
3. Using **Duality theory**
4. **Interval Analysis**

3. Using Duality Theory I

Consider the NLP Problem NLP-PR which is also termed the primal.

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x} \end{aligned} \tag{NLP-PR}$$

Definition (Lagrange function for Problem NLP-PR)

The Lagrange function (L) adds the objective function with a weighted sum of the constraint functions i.e., $L : \mathbb{R}^{n_x} \times \mathbb{R}^m \to \mathbb{R}$ is given by:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{1}$$

Each element of the vector $\boldsymbol{\lambda}$ (termed a Lagrange multiplier) corresponds to a constraint.

3. Using Duality Theory II

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Definition (The Dual function for Problem NLP-PR)

The dual function q is defined implicitly as the optimal objective function value of an optimization problem parameterized by a Lagrange multiplier vector, i.e., $q : \mathbb{R}^m \to \mathbb{R}$ is given by:

$$q(\boldsymbol{\lambda}) = \inf_{\mathbf{x} \in X} L(\mathbf{x}, \boldsymbol{\lambda}) \tag{2}$$

We note that q is only defined on $\boldsymbol{\lambda}$ for which the optimization problem above is bounded.

Definition (The Dual of Problem NLP-PR)

The Dual is an optimization problem defined as:

$$\begin{aligned} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \quad & q(\boldsymbol{\lambda}) \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}, \end{aligned} \tag{NLP-DP-DEF}$$

3. Using Duality Theory III

The above definitions are combined to write out an alternative concise Dual Problem (Problem NLP-DP)

Definition (Dual - concise definition)

$$\sup_{\boldsymbol{\lambda} \geq \mathbf{0}} \quad \inf_{\mathbf{x} \in X} f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{NLP-DP}$$

Definition (A Duality Multiplier of Problem NLP-PR)

A vector $\hat{\boldsymbol{\lambda}}$ is said to be a Duality Multiplier of Problem NLP-PR if:

$$\hat{\boldsymbol{\lambda}} \geq \mathbf{0} \quad \text{and} \quad f(\hat{\mathbf{x}}) = \inf_{\mathbf{x} \in X} f(\mathbf{x}) + (\hat{\boldsymbol{\lambda}})^{\mathrm{T}} \mathbf{g}(\mathbf{x}) \tag{3}$$

where $\hat{\mathbf{x}}$ in this case denotes the globally optimal solution. We note that the duality multiplier is not the same as a Lagrange multiplier and may not exist for certain problem classes.

3. Using Duality Theory IV

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Theorem (Weak duality)

For all problem classes, the supremum on the Dual Problem NLP-DP is a lower bound on the optimal objective function value of the primal NLP (Problem NLP-PR).

Theorem (Strong duality and existence of duality multiplier for convex programs)

Suppose Problem NLP-PR is a convex program and has a finite solution value. Also suppose a strictly feasible point exists (also called a Slater point - a feasible point where none of the constraints are active). Then strong duality holds and Problems NLP-DP and NLP-PR have the same optimum objective function value. In addition, atleast one duality multiplier exists.

Exercise : Please recall the approach Paul presented on the viewing duality through the image space

Overview

Benders' decomposition (GBD) is a technique for solving problems of the form:

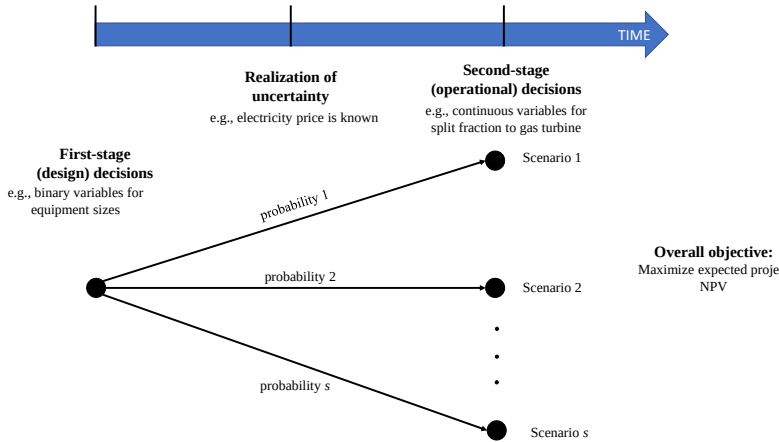
$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & f(\mathbf{x},\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{y}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x},\ \mathbf{y} \in U \subset \mathbb{R}^{n_y} \end{aligned} \tag{4}$$

where $\mathbf{y}$ is a vector of *complicating variables* in the sense that (5) is much easier to solve in $\mathbf{x}$ when $\mathbf{y}$ are held constant.

1. For constant $\mathbf{y}$, (4) separates into a number of smaller, independent problems, each involving a different subvector of $\mathbf{x}$. This can reduce running time and enables the application of parallel computing architectures. **Stochastic Programming**.
2. (4) is a not convex program in $\mathbf{x}$ and $\mathbf{y}$ jointly, but holding $\mathbf{y}$ constant renders a convex program in $\mathbf{x}$. This enables nonconvexities to be treated separately from the convex portion of the problem.

(Two-Stage) Stochastic Programming -In brief I

Formulation for Optimization under Uncertainty



(Two-Stage) Stochastic Programming -In brief II

- Sequential decision making. Variables divided into two categories: Here-and-Now, Wait-and-See. Typically in engineering: Design decisions, operational decisions. Optimize trade-offs between design and operation
- Probabilistic description of uncertainty using a scenario tree. Uncertain random variable has a known probability distribution.

Birge and Louveaux: Stochastic Programming textbook. Another excellent toolkit is Stochastic Dual Dynamic Programming (SDDP):
`https://odow.github.io/SDDP.jl/dev/tutorial/basic/01_first_steps/`

(Two-Stage) Stochastic Programming -In brief III

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Two-stage stochastic MILPs

$$
\begin{aligned}
\min_{\mathbf{y},\mathbf{x}_1,\ldots,\mathbf{x}_s} \quad & \mathbf{c}^{\mathrm{T}}\mathbf{y} + \sum_{h=1}^{s} p_h \mathbf{d}_h{}^{\mathrm{T}}\mathbf{x}_h \\
\text{s.t.} \quad & \mathbf{A}\mathbf{y} \leq \mathbf{b}, \\
& \mathbf{B}_h\mathbf{y} + \mathbf{W}_h\mathbf{x}_h \leq \mathbf{e}_h, \qquad \forall h \in \{1,\ldots,s\}, \\
& \mathbf{y} \in Y, \\
& \mathbf{x}_h \in X_h, \qquad \forall h \in \{1,\ldots,s\}
\end{aligned}
\tag{SMILP-DEP}
$$

(Two-Stage) Stochastic Programming -In brief IV

$$
\begin{pmatrix}
\mathbf{B}_1 & \mathbf{W}_1 & & & \\
\mathbf{B}_2 & & \mathbf{W}_2 & & \\
\vdots & & & \ddots & \\
\mathbf{B}_s & & & & \mathbf{W}_s
\end{pmatrix}
\begin{pmatrix}
\mathbf{y} \\
\mathbf{x}_1 \\
\mathbf{x}_2 \\
\vdots \\
\mathbf{x}_s
\end{pmatrix}
\leq
\begin{pmatrix}
\mathbf{e}_1 \\
\mathbf{e}_2 \\
\vdots \\
\mathbf{e}_s
\end{pmatrix}
$$

Benders Decomposition - Intuition I

- Problem SMILP-DEP has an almost block-diagonal structure in the sense: If the vector of variables $\mathbf{y}$ is fixed, it becomes fully decomposable into a set of linear programming (LP) problems (one for each $h \in \{1, \ldots, s\}$) .
- To partition Problem SMILP-DEP, write out an equivalent optimization problem (Problem SMILP-PART which only searches the space of the first-stage $\mathbf{y}$ variables). The objective function of Problem SMILP-PART is defined implicitly in terms of the solutions to the embedded Problems $\text{LP}_h(\mathbf{y})$ for all $h \in \{1, \ldots, s\}$.
- The set V contains all the feasible values of $\mathbf{y}$ for which all the embedded Problems $\text{LP}_h(\mathbf{y})$ have at least one feasible point.
- **Challenge:** Constructing the set V algorithmically is challenging as it would require testing all the Problems $\text{LP}_h(\mathbf{y})$ for feasibility for each candidate member $\mathbf{y}$.
- Solution: Use a duality theory-based reformulation is used to derive an equivalent set of Benders Feasibility cuts that characterizes set V.

Benders Decomposition - Intuition II

- Problems $\text{LP}_h(\mathbf{y})$ are parameterized by the particular value of the complicating variables $\mathbf{y}$. Thus, the Problems $\text{LP}_h(\mathbf{y})$ are constructed with $\mathbf{y}$ appearing on the right-hand side of the constraints (i.e., $\mathbf{y}$ is not a decision variable of Problems $\text{LP}_h(\mathbf{y})$).
- The key point is that the size of each of the Problems $\text{LP}_h(\mathbf{y})$ is independent of the number of scenarios. In addition, the problem size is smaller allowing for more efficient solution.
- Problems SMILP-DEP and SMILP-PART have the same optimal objective function value.
- The problems $\text{LP}_h(\mathbf{y})$ can be solved in parallel using a multi-core architecture.

$$\begin{aligned} \min_{\mathbf{y}} \quad & \mathbf{c}^{\mathrm{T}}\mathbf{y} + \sum_{h=1}^{s} v_h(\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{y} \in V \equiv \Big\{\mathbf{y} \in Y : \mathbf{A}\mathbf{y} \leq \mathbf{b} \text{ and } \exists \mathbf{x}_h \in X_h : \mathbf{W}_h\mathbf{x}_h \leq \mathbf{e}_h - \mathbf{B}_h\mathbf{y}, \forall h \in \{1, \ldots, s\}\Big\} \subset Y \end{aligned}$$

(SMILP-PART)

Benders Decomposition - Intuition III

$$
\begin{aligned}
v_h(\mathbf{y}) \quad = \quad & \min_{\mathbf{x}_h} \quad p_h \mathbf{d}_h{}^{\mathrm{T}} \mathbf{x}_h \\
& \text{s.t.} \quad \mathbf{W}_h \mathbf{x}_h \leq \mathbf{e}_h - \mathbf{B}_h \mathbf{y}, \\
& \qquad\quad \mathbf{x}_h \in X_h,
\end{aligned}
\tag{LP$_h(\mathbf{y})$}
$$

Summary of Strategy: Fixing the complicating first-stage $\mathbf{y}$ variables to particular values and then decomposing the problem into a number of smaller independent subproblems. Iterating between searching the space of first-stage $\mathbf{y}$ variables followed by the space of second-stage $\mathbf{x}_h$ variables.

- Benders Decomposition (BD) or the L-shaped method: LPs or MILPs. [Benders 1962]
- Generalized Benders Decomposition algorithm (GBD): MICPs. [Geoffrion 1972]
- Nonconvex Generalized Benders Decomposition algorithm (NGBD). nonconvex MINLPs, MILP recourse [Li and Barton, 2014]

Generalized Benders Decomposition - Algorithm I

Consider the problem:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & f(\mathbf{x},\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{y}) \leq \mathbf{0} \\ & \mathbf{x} \in X \subset \mathbb{R}^{n_x},\ \mathbf{y} \in U \subset \mathbb{R}^{n_y} \end{aligned} \tag{5}$$

The derivation of the Master Problem involves 3 steps:

1) Projection,
2) Dualization,
3) Relaxation

Generalized Benders Decomposition - Algorithm II

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1. Projection

Projecting (5) onto the space of complicating variables $\mathbf{y}$ to yield the *projected problem*:

$$\min_{\mathbf{y} \in V} \quad v(\mathbf{y}) \tag{6}$$

where

$$V \equiv \{\mathbf{y} \in U : \exists \mathbf{x} \in X : \mathbf{g}(\mathbf{x}, \mathbf{y}) \leq \mathbf{0}\}$$

is the set of values for $\mathbf{y}$ that are feasible. Note that this is not in general a finite set. The function $v : V \to \mathbb{R}$ is defined by

$$\forall \mathbf{y} \in V : \quad v(\mathbf{y}) \equiv \left\{ \begin{array}{ll} \inf\limits_{\mathbf{x} \in X} & f(\mathbf{x}; \mathbf{y}) \\ \text{s.t.} & \mathbf{g}(\mathbf{x}; \mathbf{y}) \leq \mathbf{0}. \end{array} \right. \qquad (\mathrm{NLP}(\mathbf{y}))$$

NLP($\mathbf{y}$) is known as the *primal problem* for $\mathbf{y}$ held constant. GBD aims to exploit the situation when the primal problem is much easier to solve than the original problem (5).

Generalized Benders Decomposition - Algorithm III

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Remarks and Reformulations:

1. The Problem NLP($\mathbf{y}$) is a convex NLP. We assume that this convex programs can either be solved to global optimality with a standard NLP solver for a given value of $\mathbf{y}$ or is infeasible. If an optimal solution $(\hat{\mathbf{x}}, \hat{\boldsymbol{\lambda}})$ is found, we assume that the NLP solver also provides the associated duality multiplier $(\hat{\boldsymbol{\lambda}})$ which is known to exist since each of the problems is convex (from Strong Duality theorem 7 Colloquium 3).
2. Problems 5 and 6 have the same objective function. However, Problem 6 is a substantially smaller problem since it only searches over the space of the $\mathbf{y}$ variables.
3. The trade-off is that the objective function of Problem 6 is defined implicitly and, for each candidate value $\mathbf{y}$, requires the solutions of the NLP($\mathbf{y}$) problem parameterized by that $\mathbf{y}$. This is computationally intractable. Instead, a duality theory-based reformulation is used as explained next:

Generalized Benders Decomposition - Algorithm IV

- First, since Problems NLP($\mathbf{y}$) are all convex, strong duality holds thus from Theorem 7, we can rewrite:

$$v(\mathbf{y}) = \sup_{\boldsymbol{\lambda} \geq \mathbf{0}} \inf_{\mathbf{x} \in X} \left\{ f(\mathbf{x}; \mathbf{y}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}; \mathbf{y}) \right\}$$

- Using the definition of the supremum, the above expression can be translated into the following form which gives an infinite number of constraints:

$$v(\mathbf{y}) \geq \inf_{\mathbf{x} \in X} \left\{ f(\mathbf{x}; \mathbf{y}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}; \mathbf{y}) \right\}, \quad \forall \boldsymbol{\lambda} \geq \mathbf{0}$$

Generalized Benders Decomposition - Algorithm V

- In the equation above, every possible vector $\boldsymbol{\lambda}$ belonging to the infinite set $\mathbb{R}_+^m$ gives rise to a new constraint. Thus, the instances of the vector $\boldsymbol{\lambda}$ can be used as an index for the constraint set, and we define a function $\mathbf{F} : U \times \mathbb{R}_+^m \to \mathbb{R}$ to represent a single one of these constraints (termed a Benders Optimality cut):

$$\mathbf{F}(\mathbf{y}, \boldsymbol{\lambda}) \equiv \inf_{\mathbf{x} \in X} f(\mathbf{x}, \mathbf{y}) + \boldsymbol{\lambda}^{\mathrm{T}} \mathbf{g}(\mathbf{x}, \mathbf{y}), \tag{7}$$

4. Another issue with Problem 6 is that testing the second membership condition of each $\mathbf{y}$ in the feasible set V is also computationally expensive. Thus, a similar reformulation is implemented to replace this condition with an (infinite) number of constraints that would exclude infeasible $\mathbf{y}$ values. Duality theory once again provides a useful result presented in the dual representation of a set theorem. We can define function:

   $\mathbf{G} : U \times M \to \mathbb{R}$ which defines a single constraint (termed a Benders Feasibility cut), with the vector $\boldsymbol{\mu}$ providing an index for the constraint set.

Generalized Benders Decomposition - Algorithm VI

5. The last reformulation of Problem 6 is for convenience: The objective function is defined in terms of a dummy decision variable $\eta \in \mathbb{R}$. Since Problem 6 is a minimization problem, the constraint below becomes active at the optimum.

$$\eta \geq v(\mathbf{y})$$

Theorem (Dual representation of second membership condition of set V)

Suppose that X is a nonempty convex set and that $g_i(\cdot, \mathbf{y})$, $i = 1, \ldots, m$ are convex on X for each fixed $\mathbf{y} \in U$. Furthermore, suppose that the set $Z(\mathbf{y}) \equiv \{\mathbf{z} \in \mathbb{R}^m : \exists \mathbf{x} \in X : \mathbf{g}(\mathbf{x}, \mathbf{y}) \leq \mathbf{z}\}$ is closed for each fixed $\mathbf{y} \in U$. Then a point $\mathbf{y} \in V$ iff $\mathbf{y} \in U$ also satisfies the infinite set of inequalities:

$$\mathbf{G}(\mathbf{y}, \boldsymbol{\mu}) \leq 0, \ \forall \boldsymbol{\mu} \in M \tag{8}$$

where

$$\mathbf{G}(\mathbf{y}, \boldsymbol{\mu}) \equiv \inf_{\mathbf{x} \in X} \boldsymbol{\mu}^{\mathrm{T}} \mathbf{g}(\mathbf{x}, \mathbf{y}) \quad \textit{and} \quad M \equiv \left\{ \boldsymbol{\mu} \in \mathbb{R}^m : \boldsymbol{\mu} \geq \mathbf{0}, \sum_{i=1}^{m} \mu_i = 1 \right\}.$$

Generalized Benders Decomposition - Algorithm VII

With the above reformulations, Problem 6 can be rewritten as Problem MP which is termed the Master Problem:

$$\begin{aligned} \min_{\eta, \mathbf{y}} \quad & \eta \\ \text{s.t.} \quad & \eta \geq \mathbf{F}(\mathbf{y}, \boldsymbol{\lambda}), \ \ \forall \boldsymbol{\lambda} \geq \mathbf{0}, \\ & \mathbf{0} \geq \mathbf{G}(\mathbf{y}, \boldsymbol{\mu}), \ \forall \boldsymbol{\mu} \in M \\ & \mathbf{y} \in Y, \eta \in \mathbb{R} \end{aligned} \tag{MP}$$

Generalized Benders Decomposition - Algorithm VIII

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Relaxed Master Problem

Problem MP is still computationally challenging to solve as it is a semi-infinite problem since it has an infinite number of constraints. This motivates the construction of a Relaxed Master Problem (Problem RMP) where only a finite subset of these constraints is iteratively added. The following features of Problem RMP are presented:

- The finite index sets T and S are introduced: Each element j of T indexes a single Benders Optimality cut $\mathbf{F}$ while each element i of S indexes a single Benders Feasibility cut $\mathbf{G}$. Thus, the elements j and s correspond to a single instance of the vectors $(\boldsymbol{\lambda})$ and $(\boldsymbol{\mu})$ respectively. At the start of the GBD Decomposition algorithm, the sets T and S are empty and one constraint is added to one of the sets at each iteration.
- Since Problem RMP is a relaxation of Problem MP, its optimal objective function value provides a rigorous lower bound (LBD) by the Lower Bounding Principle. Problem RMP gets more constrained as more cuts are added at each iteration, this LBD is guaranteed to increase or remain the same.

Generalized Benders Decomposition - Algorithm IX

- The overall strategy is to repeatedly solve Problem RMP with an increasing number of constraints. After each iteration, suppose $(\hat{\eta}, \hat{\mathbf{y}})$ is the optimal solution of Problem RMP, the next step is to test if any of the constraints left out from Problem MP are violated by this optimal solution. One crucial feature of the GBD algorithm is that this test can be performed cheaply by solving subproblems NLP($\mathbf{y}$) with $\mathbf{y}$ fixed to be $\hat{\mathbf{y}}$. If the optimal solutions for all the NLP($\mathbf{y}$) subproblems are found and $\hat{\eta} < v(\hat{\mathbf{y}})$, then it can be shown that (at least) one of the constraints of Problem MP has been violated. The intuition for the above test is as follows: If $(\hat{\eta}, \hat{\mathbf{y}})$ did not violate any constraints of Problem MP, then as a result of strong duality, $v(\hat{\mathbf{y}})$ should be equal to $\hat{\eta}$. Thus, failure of this test implies that a Benders Optimality cut should be added to Problem RMP.
- **Generating new optimality cuts:** Use duality multipliers! If the NLP solver for subproblems NLP($\mathbf{y}$) provides duality multipliers $(\hat{\boldsymbol{\lambda}})$ (which exist for convex programs from Theorem 7): The infinite number of constraints in the master problem are indexed by the vectors $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\boldsymbol{\mu} \in M$. From $\hat{\eta} < v(\hat{\mathbf{y}})$, we see that any $\hat{\boldsymbol{\lambda}} \geq \mathbf{0}$ such that

$$\hat{\eta} < \inf_{\mathbf{x} \in X} f(\mathbf{x}, \hat{\mathbf{y}}) + \hat{\boldsymbol{\lambda}}^{\mathrm{T}} \mathbf{g}(\mathbf{x}, \hat{\mathbf{y}})$$

Generalized Benders Decomposition - Algorithm X

gives an index of a violated constraint. A duality multiplier $\hat{\boldsymbol{\lambda}}$ for NLP($\hat{\mathbf{y}}$) is just such an index because by the definition of the duality multiplier:

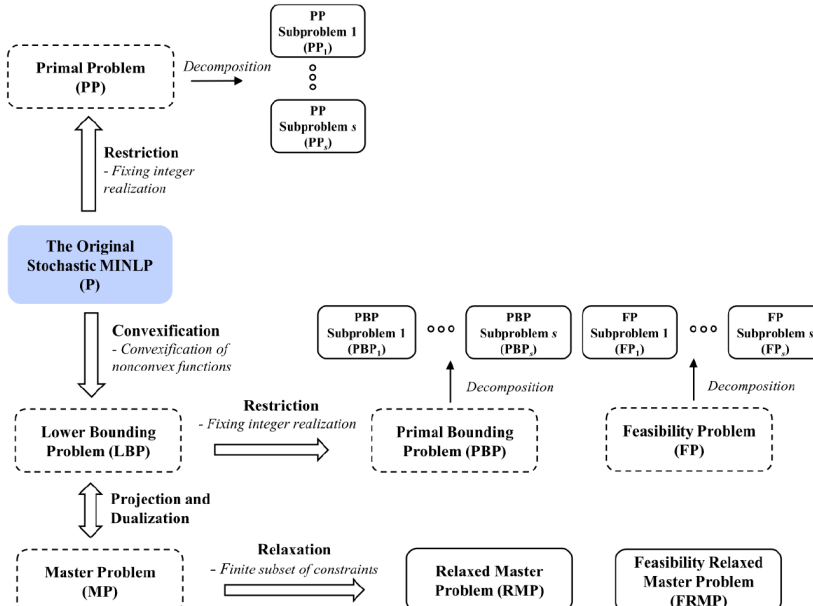
$$\hat{\eta} < v(\hat{\mathbf{y}}) = \inf_{\mathbf{x} \in X} f(\mathbf{x}, \hat{\mathbf{y}}) + \hat{\boldsymbol{\lambda}}^{\mathrm{T}} \mathbf{g}(\mathbf{x}, \hat{\mathbf{y}}).$$

Most modern convex NLP algorithms will generate a duality multiplier as part of the solution procedure, if one exists..

- Conversely, if any of Problems NLP($\mathbf{y}$) (with $\mathbf{y}$ fixed to be $\hat{\mathbf{y}}$) is infeasible, then $\hat{\mathbf{y}}$ violates one of the Benders Feasibility cuts of Problem MP. Without proof, we mention that the set of duality multipliers obtained from solution of the feasibility problems provides an index for one violated Benders Feasibility cut.
- In summary, at each iteration either a Benders Optimality or Feasibility cut is added to the Relaxed Master Problem by appending a set of duality multipliers to either set T or S obtained from the PBP or the FP.

Generalized Benders Decomposition - Algorithm XI

GBD - Summary

1. Begin with initial guess for the $\mathbf{y}$ variables. Solve the Primal problem NLP($\mathbf{y}$).
2. If this problem is feasible, an UBD is attained. Use the duality multiplier from the solver of NLP($\mathbf{y}$) to generate a Benders Optimality cut for the Relaxed Master Problem.
3. If the primal problem is infeasible, a Feasibility Problem is solved and the duality multipliers attained are used to generate Benders Feasibility cuts for the Relaxed Master Problem.
4. The Relaxed Master Problem is derived by relaxing the constraints of the Master Problem.
5. This Master Problem in turn is derived by partitioning (projection) of the convex problem onto the space of the $\mathbf{y}$ variables, and replacing the embedded optimization problem (in the $\mathbf{x}$ space) with an infinite number of constraints by leveraging its dual representation. Similarly, a dual representation is utilized to define the set of feasible $\mathbf{y}$ values.
6. The Relaxed Master Problem provides a lower bound and its solution gives the $\mathbf{y}$ value used for the next iteration of the Primal Bounding Problems.
7. The procedure is iterated until convergence

(Nonconvex) Generalized benders Decomposition - Algorithm